

C L E C S

Particle physics involves the study of the most fundamental objects and forces in nature. In the 20th century, the discovery of a “zoo” of elementary and composite particles allowed scientists to build the most successful theory yet for understanding particle physics: the Standard Model. This proposes that matter is made up of 12 fundamental particles (fermions), while the three quantum forces—strong, weak, and electromagnetic—are carried by force-carrying particles (bosons). According to quantum field theory, all these particles are excitations of their underlying quantum fields.